TWO KISSING CONFIGURATIONS OF 8358 POINTS IN EIGHTEEN DIMENSIONS

Abstract. We give two kissing configurations of 8358 points in 18 dimensions. This exceeds the best published lower bound, 7654 by Cohn and Li, by 704, and ties a value reported earlier by Kravatskiy in a public repository. The first configuration is a hyperplane section: we cut the base of Ho's 19-dimensional kissing configuration by the hyperplane orthogonal to the only word of weight 3 in his code, and project his sign vectors into it. The section keeps 7398 of the base vectors of squared norm 8, the same number as in Leech's 18-dimensional configuration, and three quarters of Ho's 1280 sign vectors survive the projection, because it lowers their squared norm from 19 to 56/3 while Cohn and Li's argument needs only 18. Seen in coordinates adapted to the hyperplane, the result has a simple shape: a 16-dimensional head built from the Reed–Muller codes of length 16, a regular hexagon of six directions in a two-dimensional tail, the sextads of a bent quadratic form attached to each direction, and the added points at the six midpoint angles. The second configuration fills the same shape with six different bent forms. Both proofs are by hand. The two configurations are not isometric: the first contains the antipodes of at least 7398 of its points, the second of exactly 4326.

1. Introduction

The kissing problem in $\mathbb{R}^n$ asks how many unit spheres can touch a central unit sphere without overlapping [3, Ch. 1, §2.2]. In terms of the points of tangency, it asks for the maximum size τ_n of a set X of unit vectors in $\mathbb{R}^n$ with $\langle x, y\rangle \le 1/2$ for all distinct $x, y \in X$. The value of τ_n is known only for $n \in \{1, 2, 3, 4, 8, 24\}$; see [2, Table 1.1] for references. For $17 \le n \le 23$ the best constructions were Leech's cross sections of the Leech lattice minimal vectors [6] until Cohn and Li [2] improved dimensions 17 through 21 by changing the signs in those constructions. In dimension 18 they obtain $\tau_{18} \ge 7654$. Ho [4] then raised the bound in dimension 19 to 11948 with a larger code.

Theorem 1.1. $\tau_{18} \ge 8358$. *There are two non-isometric kissing configurations of 8358 points in* $\mathbb{R}^{18}$.

Table 1 puts this value in context. Kravatskiy's public repository [5] reported 8358 points in dimension 18 before us, by a construction built from bent cosets of a Reed–Muller code, which has the same shape as ours (Section 7). Our value is a tie with the repository value and an improvement of 704 over the best published bound.

Table 1. Lower and upper bounds for τ_{18}. The published lower bound and the upper bound are from Cohn's table [1]; the upper bound is the floor of a numerical semidefinite programming value [7] and is not interval-certified. The repository value is from [5].

dimension	published lower bound	repository	this paper	upper bound
18	7654 [2]	8358 [5]	8358	16406 [7]

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The idea. Cohn and Li [2, §2] build kissing configurations in $\mathbb{R}^n$ from a base of vectors of squared norm 8, namely the roots $\pm 2e_i \pm 2e_j$ and the vectors with entries ± 1 on an octad of a code and an odd number of minus signs, and then add sign vectors $q \in \{\pm 1\}^n$ of words orthogonal to the octads. A parity argument gives $\langle b, q\rangle \le 6$ for every base vector b, so q is compatible with the base as soon as $6 \le \frac{1}{2}\sqrt{8|q|^2}$, that is, $|q|^2 \ge 18$. In 19 dimensions $|q|^2 = 19$, and there is one unit to spare. Ho [4] uses this room only through the code: he finds a nonlinear code of 1280 words at pairwise distance at least 5, and adds their sign vectors to Cohn and Li's base of 10668 vectors.

We spend part of the spare unit to lose a dimension. Let s be the indicator vector of the only word of weight 3 in Ho's code, and project orthogonally onto the hyperplane $s^\perp$. A base vector orthogonal to s keeps its inner product with every sign vector. A sign vector q with $\langle q, s\rangle = \pm 1$ loses only 1/3 of its squared norm, leaving $56/3 > 18$, so it still fits; the 320 sign vectors with $\langle q, s\rangle = \pm 3$ lose 3, leaving 16, and do not fit. The base vectors orthogonal to s number 7398, which is Leech's count in 18 dimensions and the number from which Cohn and Li start there. They then add 256 points. The projection adds 960.

Sections 2 to 4 give the section and show that the tail of each added point can be rescaled within an interval. Section 5 describes the shape of the result in coordinates adapted to the hyperplane: a 16-dimensional head carrying the Reed–Muller codes of length 16, a two-dimensional tail with a regular hexagon of six directions, the sextads of a bent quadratic form on $\mathbb{F}_2^4$ attached to each direction, and the added points at the six midpoint angles. We prove that this shape gives a kissing configuration under three conditions on the forms, and that the section is an instance of it, with three families from the Golay code, each serving two opposite directions. Section 6 fills the shape in a second way, with six different forms, and reaches 8358 again without passing through 19 dimensions. Section 7 proves that the two configurations are not isometric and compares them with Kravatskiy's, which fills the same shape in a third way. The two main theorems are proved by hand; the computed facts are named where they are used and listed in Section 8. Coordinates of the three configurations and a program that checks every pair in exact arithmetic accompany the paper.

Notation. Coordinates of $\mathbb{R}^{19}$ are numbered $1, \dots, 19$, and

$$T = \{2, 11, 14\}, \qquad H = \{1, \dots, 19\} \setminus T, \qquad s = \mathbf{1}_T = e_2 + e_{11} + e_{14}.$$

We call H the *head* and T the *tail*. For $x \in \mathbb{R}^{19}$ we write x_H and x_T for the vectors that agree with x on H and on T and vanish elsewhere, so $x = x_H + x_T$. The hyperplane $s^\perp$ is our copy of $\mathbb{R}^{18}$, with the inner product of $\mathbb{R}^{19}$.

For nonzero x, y the angle between them is at least 60° if and only if

$$\langle x, y\rangle \le 0 \quad \text{or} \quad 4\langle x, y\rangle^2 \le |x|^2|y|^2. \tag{1.1}$$

We call such a pair *compatible*. If every pair from a finite set of nonzero vectors is compatible, the normalized vectors form a kissing configuration; distinct vectors give distinct points, because (1.1) excludes cosine 1. A pair with $\langle x, y\rangle = \frac{1}{2}|x|\,|y|$ is a *contact*.

2. Ho's configuration

We identify a binary word with its support. For $c \in \mathbb{F}_2^{19}$ we write $q(c) = (-1)^c \in \{\pm 1\}^{19}$ for its sign vector. If c, c' are at Hamming distance d, then $\langle q(c), q(c')\rangle = 19 - 2d$.

The codes. Let $D \le \mathbb{F}_2^{19}$ be the linear code spanned by Ho's twelve words [4, §2]

$$\begin{aligned} m_1 &= \{1,8,9,12,16,17,18,19\}, & m_2 &= \{2,10,11,14,15,17,18\},\\ m_3 &= \{3,7,9,13,15,16,17,19\}, & m_4 &= \{4,7,8,10,12,15,16,19\},\\ m_5 &= \{5,10,12,13,15,16,17,18\}, & m_6 &= \{6,7,8,9,10,13,16,18\},\\ a_1 &= \{1,4,7,9\}, & a_2 &= \{1,5,6,18\},\\ a_3 &= \{1,3,12,15\}, & a_4 &= \{1,10,13,19\},\\ r_1 &= \{1,3,5,6,7,13,14,15,18\}, & r_2 &= \{2,4,6,7,8,13,14,16,17,18\}. \end{aligned}$$

Let $M = \langle m_1, \dots, m_6 \rangle$, let $a_5 = \{3,5,7,10\} = a_1 + a_2 + a_3 + a_4 + m_4 + m_6$, and let $R = \{0, r_1, r_2, r_1 + r_2\}$. Ho's code is

$$A = \bigcup_{k=1}^{5} \bigcup_{r \in R} (M + a_k + r) \ \subseteq D,$$

the code of [4, Proposition 9]; it has 1280 words, and distinct words are at Hamming distance at least 5.

Ho proves [4, Proposition 2] that D is the extended binary Golay code G on 24 coordinates, restricted to the coordinates $1, \dots, 19$; in other words, D is G punctured at the five coordinates $P = \{20, \dots, 24\}$. We use the standard facts about G [3, Ch. 10, §2]: its nonzero weights are 8, 12, 16 and 24; its 759 words of weight 8, the octads, form a Steiner system $S(5,8,24)$, so every five coordinates lie in exactly one octad; two octads meet in 0, 2 or 4 coordinates; and for a fixed octad O, exactly 30 octads are disjoint from O and exactly 16 meet O in any given pair of its points.

Lemma 2.1. (1) *$T = \{2,11,14\}$ is the only word of weight 3 in D.*

(2) *$D^\perp$ consists of the words of G that vanish on P. Its words of weight 8, which we also call* octads*, are the 78 octads of G disjoint from P. Two distinct octads of $D^\perp$ meet in 0, 2 or 4 coordinates.*

(3) *Every octad of $D^\perp$ meets T in 0 or 2 coordinates. Exactly 30 are disjoint from T, and for each 2-subset of T exactly 16 meet T in that subset.*

Proof. (1) A word of weight 3 in D is the restriction of a word $c \in G$ with $\mathrm{wt}(c) - |c \cap P| = 3$. Since $|P| = 5$ and the weights of G are $0, 8, 12, 16, 24$, this forces $\mathrm{wt}(c) = 8$ and $P \subseteq c$. So c is the unique octad O through P, and $T = O \setminus P$. In Ho's coordinates $T = m_1 + m_2 + m_4 + a_1 = \{2,11,14\}$.

(2) The dual of a punctured code is the shortened code [8]: since G is self-dual, $D^\perp$ is the set of words of G supported on $\{1, \dots, 19\}$, restricted to those coordinates. Its words of weight 8 are the octads of G disjoint from P, and they meet as octads of G do.

(3) An octad o of $D^\perp$ is disjoint from P, so $o \cap O = o \cap T$, which has 0, 2 or 4 elements; as $|T| = 3$, it has 0 or 2. The 30 octads disjoint from O are disjoint from P, and so are the 16 octads through a given pair of T that meet O in exactly that pair. These $30 + 3 \cdot 16 = 78$ octads are all of them. □

The base and the parity lemma. Let B be the set of the $4\binom{19}{2} = 684$ *roots* $\pm 2e_i \pm 2e_j$ together with, for each octad o of $D^\perp$, the 128 vectors with entries ± 1 on o, zeros elsewhere and an odd number of minus signs. These $78 \cdot 128 = 9984$ vectors are the *octad vectors*, and the 128 on o form the *fiber* over o. All 10668 vectors of B have squared norm 8. This is the odd-sign configuration of Cohn and Li [2, §2] for the 78 octads, their 19-dimensional base. Ho's configuration is B together with the 1280 vectors $\sqrt{8/19}\, q(c)$, $c \in A$.

Lemma 2.2 (Cohn and Li)**.** *Distinct $b, b' \in B$ satisfy $\langle b, b' \rangle \leq 4$.*

Proof. Two distinct roots have inner product in $\{0, \pm 4, -8\}$. A root has two nonzero coordinates, each contributing at most 2 against an octad vector. Two vectors on the same octad have odd sign patterns, so they differ in an even positive number d of places and have inner product $8 - 2d \leq 4$. Two vectors on different octads share at most four coordinates by Lemma 2.1(2). □

Lemma 2.3 (Cohn and Li)**.** *Let $c \in D$ and $q = q(c)$. For an octad vector b we have $\langle b, q \rangle \leq 6$, and for a root b we have $\langle b, q \rangle \leq 4$.*

Proof. Let b lie on the octad o. Since $o \in D^\perp$, the word c meets o in an even number of coordinates. So q has an even number of minus signs on o, while b has an odd number. Hence q and b disagree in an odd number $m \geq 1$ of coordinates of o, and $\langle b, q \rangle = 8 - 2m \leq 6$. A root has two entries ± 2, so $\langle b, q \rangle \leq 4$. □

For $c \in D$ the sign vector $q(c)$ has squared norm 19 and $\frac{1}{2}\sqrt{8 \cdot 19} > 6$, so it is compatible with B. Two sign vectors of words at distance $d \geq 5$ have inner product $19 - 2d \leq 9 < 19/2$. This is Ho's configuration of $10668 + 1280 = 11948$ points.

3. The section

The body. Let $B_s = \{b \in B : \langle b, s \rangle = 0\}$. We sort its vectors into four types: head roots, tail roots, full fibers and half fibers.

Lemma 3.1. *$|B_s| = 486 + 3840 + 3072 = 7398$, and B_s spans $s^\perp$.*

Proof. Head roots. A root $\pm 2e_i \pm 2e_j$ with $i, j \in H$ lies in $s^\perp$; there are $4\binom{16}{2} = 480$.

Tail roots. A root with exactly one index in T has $\langle b, s \rangle = \pm 2$, so it is not in B_s. A root with both indices in T lies in $s^\perp$ exactly when its two signs are opposite. This gives the 6 roots $2(e_i - e_j)$, $i \neq j \in T$. In all there are $480 + 6 = 486$ roots.

Full fibers. If an octad is disjoint from T, its whole fiber lies in $s^\perp$. By Lemma 2.1(3) these are $30 \cdot 128 = 3840$ vectors.

Half fibers. Let $o \cap T = \{i, j\}$. An octad vector b on o has $\langle b, s \rangle = b_i + b_j$, which vanishes exactly when $b_j = -b_i$. Then b_i, b_j contribute exactly one minus sign, so the six entries on $o \setminus T$ must have an even number of minus signs. This gives $2 \cdot 2^5 = 64$ vectors, half the fiber. By Lemma 2.1(3) these are $48 \cdot 64 = 3072$ vectors.

Span. The head roots span $\mathbb{R}^H$, and the tail roots span the plane $\{x \in \mathbb{R}^T : x_2 + x_{11} + x_{14} = 0\}$. Together they span $s^\perp$. □

The count 7398 is the size of Leech's 18-dimensional configuration, and of the configuration from which Cohn and Li start in 18 dimensions [2, §4].

The projected sign vectors. For $c \in \mathbb{F}_2^{19}$ put $q = q(c)$ and $h(c) = \langle q, s \rangle = q_2 + q_{11} + q_{14} \in \{\pm 1, \pm 3\}$. Let π be the orthogonal projection onto $s^\perp$, so that

$$\pi(q) = q - \tfrac{h}{3}s, \qquad |\pi(q)|^2 = 19 - \tfrac{h^2}{3}, \qquad \langle b, \pi(q) \rangle = \langle b, q \rangle \ \text{ for } b \in s^\perp.$$

If $h = \pm 3$, then $|\pi(q)|^2 = 16 < 18$, and an octad vector b in $s^\perp$ with $\langle b, q \rangle = 6$ (there is one for every full fiber, by the proof of Lemma 2.3) has cosine $6/\sqrt{8 \cdot 16} > \frac{1}{2}$ with $\pi(q)$. So only the words with $h = \pm 1$ can be used, and for them $|\pi(q)|^2 = 56/3 > 18$. For $h = \pm 3$ the tail $q_T - \frac{h}{3}s$ of the projection is zero, so no rescaling of the tail, as in Section 4, changes the squared norm 16.

Lemma 3.2. *Let $A_1 = \{c \in A : h(c) = \pm 1\}$. Then*

$$A_1 = \bigcup_{k=1}^{5} \bigcup_{r \in \{r_1, r_2, r_1+r_2\}} (M + a_k + r), \qquad |A_1| = 960.$$

The restriction $c \cap T$ of a word $c \in A_1$ takes each of the six values $\{14\}$, $\{2, 11\}$, $\{2, 14\}$, $\{11\}$, $\{2\}$, $\{11, 14\}$ for exactly 160 words. The 480 words with a one-element restriction have $h = 1$, and the 480 with a two-element restriction have $h = -1$.

Proof. We read off the generators. Among $m_1, \dots, m_6$ only m_2 meets T, and $m_2 \supseteq T$; so the restrictions to T of the words of M are exactly $\emptyset$ and T, each for 32 words. None of $a_1, \dots, a_5$ meets T. Finally $r_1 \cap T = \{14\}$, $r_2 \cap T = \{2, 14\}$ and $(r_1 + r_2) \cap T = \{2\}$. Hence a word of $M + a_k + r$ has restriction $r \cap T$ or $(r \cap T) + T$, each for 32 words.

For $r = 0$ the restriction is $\emptyset$ or T, so $h = \pm 3$ and the word is not in A_1. For $r \neq 0$ the restriction has one or two elements, so $h = \pm 1$. This gives $5 \cdot 3 \cdot 64 = 960$ words, 160 for each of the six restrictions. A one-element restriction has one minus sign among q_2, q_{11}, q_{14}, so $h = 1$; a two-element restriction has $h = -1$. □

For $c \in A_1$ we use the integer vector

$$y(c) = 3\pi(q(c)) = 3q(c) - h(c)\, s, \qquad |y(c)|^2 = 9 \cdot \tfrac{56}{3} = 168.$$

Theorem 3.3. *The 7398 vectors of B_s and the 960 vectors $y(c)$, $c \in A_1$, are pairwise compatible. Their normalizations form a kissing configuration of 8358 points in $s^\perp \cong \mathbb{R}^{18}$.*

Proof. There are three kinds of pair.

Body and body. By Lemma 2.2, $\langle b, b' \rangle \leq 4 = \frac{1}{2} \cdot 8$.

Body and added. For $b \in B_s$ we have $\langle b, y(c) \rangle = 3\langle b, q(c) \rangle \leq 18$ by Lemma 2.3, and $4 \cdot 18^2 = 1296 < 1344 = 8 \cdot 168$.

Added and added. Let $c \neq c'$ in A_1 be at Hamming distance d, so that $\langle q(c), q(c') \rangle = 19 - 2d$. Since $\langle \pi(q), \pi(q') \rangle = \langle q, q' \rangle - \frac{1}{3} h h'$,

$$\langle y(c), y(c') \rangle = 9(19 - 2d) - 3h(c)h(c').$$

As $d \geq 5$ and $h(c)h(c') = \pm 1$, this is at most $81 + 3 = 84 = \frac{1}{2} \cdot 168$.

The count is Lemmas 3.1 and 3.2, and the vectors span $s^\perp$ by Lemma 3.1. □

This proves the bound of Theorem 1.1; the second configuration is in Section 6 and the non-isometry in Section 7. The cosine of a body vector and an added vector equals $18/\sqrt{8 \cdot 168} = 3\sqrt{21}/28$ for every full-fiber vector b with $\langle b, q(c) \rangle = 6$, and such a b exists for every c by the proof of Lemma 2.3. This value does not occur among the Leech lattice minimal vectors, whose cosines lie in $\{0, \pm\frac{1}{4}, \pm\frac{1}{2}, \pm 1\}$, so the configuration is not a cross section of them.

4. Slack

Each added vector splits into a head and a tail: $\pi(q) = q_H + t(c)$ with

$$t(c) = \pi(q)_T = q_T - \tfrac{h}{3} s.$$

We now show that the tail of every added point can be rescaled by its own factor. The lower limit of the allowed interval is exactly Cohn and Li's condition $|q|^2 \geq 18$.

Lemma 4.1. *Let $c, c' \in A_1$.*

(1) *$|t(c)|^2 = 8/3$, and up to a permutation of the three tail coordinates $t(c) = \pm\frac{1}{3}(2, 2, -4)$. The six values of t are the vertices of a regular hexagon in the plane $s^\perp \cap \mathbb{R}^T$, each taken by 160 words.*

(2) $\langle t(c), t(c')\rangle$ *is* $\frac{8}{3}$, $\frac{4}{3}$, $-\frac{4}{3}$ *or* $-\frac{8}{3}$ *according as the restrictions* $c\cap T$ *and* $c'\cap T$ *differ in 0, 1, 2 or 3 tail coordinates. We call tails with inner product* $\frac{4}{3}$ adjacent.

(3) *Adjacent tails have opposite values of* h.

Proof. (1) $|t(c)|^2 = 3 - \frac{2}{3}h^2 + \frac{1}{3}h^2 = \frac{8}{3}$ since $h^2 = 1$. If $c\cap T = \{14\}$ then $q_T = (1,1,-1)$ on $(2,11,14)$, $h = 1$ and $t = \frac{1}{3}(2,2,-4)$; the other five cases are permutations and negations of this one, and the six restrictions of Lemma 3.2 give the six vectors.

(2) $\langle t, t'\rangle = \langle q_T, q'_T\rangle - \frac{1}{3}hh'$, and $\langle q_T, q'_T\rangle = 3 - 2e$ where e is the number of tail coordinates in which the restrictions differ. If $e \in \{0,2\}$ the two restrictions have the same size parity, so $hh' = 1$ and the product is $\frac{8}{3} - 2e \in \{\frac{8}{3}, -\frac{4}{3}\}$. If $e \in \{1,3\}$ then $hh' = -1$ and the product is $\frac{10}{3} - 2e \in \{\frac{4}{3}, -\frac{8}{3}\}$.

(3) is the case $e = 1$. □

Lemma 4.2 (head distances, computed)**.** *Let* $c \neq c'$ *in* A_1*, with* d_H *the number of head coordinates in which they differ. If* $t(c) = t(c')$*, then* $d_H \geq 6$*; otherwise* $d_H \geq 4$*. Exactly 13056 unordered pairs have adjacent tails and* $d_H = 4$.

Proof. A direct check of the $\binom{960}{2}$ pairs in exact integers. The distance 5 of A alone gives only $d_H \geq 5 - e$ for tails that differ in e coordinates; the lemma is stronger for $e \neq 1$. □

Theorem 4.3. *For each* $c \in A_1$ *choose* $\mu_c > 0$ *and put* $x(c) = q(c)_H + \mu_c\, t(c)$*, so that* $N_c := |x(c)|^2 = 16 + \frac{8}{3}\mu_c^2$*. If*

$$18 \;\leq\; N_c \;\leq\; 176 - 64\sqrt{6} \approx 19.23 \qquad \text{for all } c \in A_1,$$

then the 7398 vectors of B_s *and the 960 vectors* $x(c)$ *are pairwise compatible, so their normalizations form a kissing configuration of 8358 points in* $s^\perp$*. The choice* $\mu_c = 1$*,* $N_c = 56/3$*, is Theorem 3.3.*

Proof. Write $q = q(c)$, $t = t(c)$ and $\mu = \mu_c$. The condition on N_c is $\frac{3}{4} \leq \mu^2 \leq 60 - 24\sqrt{6}$, that is, $\frac{\sqrt{3}}{2} \leq \mu \leq 6 - 2\sqrt{6}$.

Body and added. Let $b \in B_s$. The pair is compatible when $\langle b, x(c)\rangle \leq \frac{1}{2}\sqrt{8N_c} = \sqrt{2N_c}$. Put $\alpha = \langle b_H, q_H\rangle$ and $\delta = \langle b_T, q_T\rangle$. Since $\langle b_T, s\rangle = \langle b, s\rangle = 0$ we have $\langle b_T, t\rangle = \delta$, and so $\langle b, x(c)\rangle = \alpha + \mu\delta$. We bound α and δ for each of the four types of b.

Head roots. $\delta = 0$ and $\alpha \leq 4$ by Lemma 2.3, and $4 < \sqrt{2N_c}$ always.

Tail roots. $b = 2\varepsilon(e_i - e_j)$ with $i, j \in T$, so $\alpha = 0$ and $\delta = 2\varepsilon(q_i - q_j) \leq 4$. Now $4\mu \leq \sqrt{2N_c}$ is equivalent to $\mu^2 \leq 3$.

Full fibers. $\delta = 0$ and $\alpha = \langle b, q\rangle \leq 6$ by Lemma 2.3. Now $6 \leq \sqrt{2N_c}$ is equivalent to $N_c \geq 18$, with equality only at the lower end.

Half fibers. Let b lie on an octad o with $o \cap T = \{i, j\}$. Then $b_T = \varepsilon(e_i - e_j)$, so $\delta = \varepsilon(q_i - q_j) \in \{0, \pm 2\}$. Lemma 2.3 gives $\alpha + \delta = \langle b, q\rangle \leq 6$, and $\alpha \leq 6$ because b_H has six nonzero entries. If $\delta = 0$, then $\langle b, x(c)\rangle \leq 6$ as for full fibers. If $\delta = -2$, then $\langle b, x(c)\rangle \leq 6 - 2\mu < 6$. If $\delta = 2$, then $\alpha \leq 4$ and $\langle b, x(c)\rangle \leq 4 + 2\mu$, and

$$(4 + 2\mu)^2 \leq 2N_c = 32 + \tfrac{16}{3}\mu^2 \iff \mu^2 - 12\mu + 12 \geq 0,$$

which holds for $0 < \mu \leq 6 - 2\sqrt{6}$, with equality only at the upper end.

Added and added. Let $c \neq c'$ with factors μ, μ', and write $q' = q(c')$, $t' = t(c')$. Then $\langle x(c), x(c')\rangle = \langle q_H, q'_H\rangle + \mu\mu'\langle t, t'\rangle$, with $\langle q_H, q'_H\rangle = 16 - 2d_H$, and the pair is compatible when this is at most $\frac{1}{2}\sqrt{N_c N_{c'}} = \frac{4}{3}\sqrt{(6 + \mu^2)(6 + \mu'^2)}$, which is at least 9. By Lemma 4.1(2) there are three cases.

- Equal tails, $\langle t, t' \rangle = \frac{8}{3}$. Lemma 4.2 gives $d_H \geq 6$, so $\langle x(c), x(c') \rangle \leq 4 + \frac{8}{3}\mu\mu' \leq 4 + \frac{8}{3}(60 - 24\sqrt{6}) < 8$.
- Adjacent tails, $\langle t, t' \rangle = \frac{4}{3}$. Lemma 4.2 gives $d_H \geq 4$, so

$$\langle x(c), x(c') \rangle \leq 8 + \tfrac{4}{3}\mu\mu' = \tfrac{4}{3}\big(\sqrt{6} \cdot \sqrt{6} + \mu\mu'\big) \leq \tfrac{4}{3}\sqrt{(6 + \mu^2)(6 + \mu'^2)}$$

  by the Cauchy–Schwarz inequality for the vectors $(\sqrt{6}, \mu)$ and $(\sqrt{6}, \mu')$, with equality only if $\mu = \mu'$ and $d_H = 4$.
- Otherwise $\langle t, t' \rangle < 0$, so $\langle x(c), x(c') \rangle < 8$.

Body pairs are compatible by Lemma 2.2. □

Remark 4.4 (contacts). An added point touches another point only in three cases: at the lower end $N_c = 18$, against the octad vectors b with $\alpha = 6$, $\delta = 0$; at the upper end $N_c = 176 - 64\sqrt{6}$, against the half-fiber vectors with $\delta = 2$, $\alpha = 4$; and against $x(c')$ when the tails are adjacent, $d_H = 4$ and $\mu_c = \mu_{c'}$. For the configuration of Theorem 3.3, where all $\mu_c = 1$, the added points therefore touch only each other, in the 13056 pairs of Lemma 4.2; the body has 1 354 374 contacts (computer count). If the N_c are chosen in the open interval and $\mu_c \neq \mu_{c'}$ whenever the tails of c and c' are adjacent, no added point touches anything. Each added point can then be moved a little in any direction, independently of the others, so the configuration is far from rigid. None of these motions makes room for an 8359th point, as far as we know.

Remark 4.5 (a spherical 3-design). Take $\mu_c = \beta$ for all c, with $\beta^2 = 174/151$; then $N_c = 2880/151 \approx 19.07$ lies in the interval of Theorem 4.3. The resulting 8358 unit vectors in $s^\perp$ form a spherical 3-design, that is, their first and third moments vanish and $\sum_x xx^\mathsf{T} = \frac{8358}{18} I$ on $s^\perp$. The body is antipodal, so it contributes nothing to the odd moments; the odd moments of the added points and the head and mixed blocks of the second moment vanish by a computer check. The tail block is a short calculation, and it is where β comes from. A regular m-gon of vectors of squared radius ρ in a plane has $\sum vv^\mathsf{T} = \frac{m\rho}{2}$ times the projection P_T onto that plane. The six tail roots (squared radius 8) give $24P_T$; the tails $\pm(e_i - e_j)$ of the half-fiber vectors, 512 copies of each (squared radius 2), give $3072P_T$; and the 960 tails $\beta t(c)$, 160 hexagons of squared radius $\frac{8}{3}\beta^2$, give $1280\beta^2 P_T$. Dividing by the squared norms 8 and N_c,

$$\frac{24 + 3072}{8} P_T + \frac{151}{2880} \cdot 1280\beta^2 P_T = \Big(387 + \frac{1392}{18}\Big) P_T = \frac{8358}{18} P_T.$$

No added point of this design has its antipode in the configuration: $-x(c)$ has sixteen nonzero head entries, so it is not a body vector, and $-x(c) = x(c')$ would force $q(c') = -q(c)$, while no two words of A_1 are complementary (computer check).

5. The shape of the configuration

The section of Ho's configuration is easier to see in coordinates adapted to the hyperplane. Its tail, the plane $s^\perp \cap \mathbb{R}^T$, contains a regular hexagon: the six tail roots sit at the vertices, the tails $\pm(e_i - e_j)$ of the half-fiber vectors point along the same six directions, and the six tails $t(c)$ of the added points sit at the midpoint angles (Lemma 4.1). Its head carries the Reed–Muller codes of length 16, and the six-element supports of the half-fiber vectors are the sextads of bent quadratic forms. This section states the shape precisely and proves that it gives a kissing configuration whenever a few conditions on the forms hold. Section 6 fills the shape with other forms, and Section 7 shows that Kravatskiy's configuration has the same shape again.

Bent quadratic forms. Index the 16 head coordinates by the points $x = (x_0, x_1, x_2, x_3)$ of $\mathbb{F}_2^4$, and identify a Boolean function on $\mathbb{F}_2^4$ with its support, a word of length 16. The 32 affine functions form the Reed–Muller code $\mathrm{RM}(1,4)$; its 30 nonconstant words have weight 8, and we call their supports *octads*, since two of them meet in 0 or 4 points. The functions of degree at most 2 form $\mathrm{RM}(2,4)$, the dual of $\mathrm{RM}(1,4)$, with minimum distance 4. A homogeneous quadratic form $q = \sum_{i<j} a_{ij} x_i x_j$ is *bent* if its 32 translates $q + \ell$, ℓ affine, all have weight 6 or 10; this holds exactly when the alternating matrix (a_{ij}) is nonsingular over $\mathbb{F}_2$, and then 16 translates have weight 6 and 16 have weight 10 [8, Ch. 14]. We call the 16 supports of weight 6 in $q + \mathrm{RM}(1,4)$ the *sextads* of q, and the set of them the *bent family* of q. Two sextads of the same form differ by an affine word of weight 8, so they meet in 2 points. If $q + q'$ is bent, a sextad of q and a sextad of q' differ by a word of weight 6 or 10, so they meet in 3 or 1 points. An octad and a sextad of q differ by a word of $q + \mathrm{RM}(1,4)$, so they meet in 4 or 2 points.

For two Boolean functions f, h on $\mathbb{F}_2^4$, the number $|\operatorname{supp} f \cap \operatorname{supp} h|$ is congruent modulo 2 to the coefficient of $x_0x_1x_2x_3$ in fh, because every other monomial has even weight. If $f = q + \ell$ and $h = q' + \ell'$ with q, q' quadratic and ℓ, ℓ' affine, this coefficient is the coefficient of $x_0x_1x_2x_3$ in qq', since the other products have degree at most 3. We write $\langle q, q' \rangle_4$ for it.

Hexagon configurations. Let $u_\theta = (\cos\theta, \sin\theta) \in \mathbb{R}^2$, write $u_k = u_{k\pi/3}$ for the six hexagon directions, and $\theta_j = (j + \frac{1}{2})\pi/3$ for the six midpoint angles, so that the sector j lies between the directions u_j and u_{j+1} (indices modulo 6). The data of a *hexagon configuration* are six bent forms $q_0, \dots, q_5$, one for each direction, and six sets $G_0, \dots, G_5$ of quadratic forms, one for each sector, such that

- (H1) for $k \neq k'$, either $q_k + q_{k'}$ is bent, or $u_{k'} = -u_k$ and $q_{k'} = q_k$;
- (H2) in each G_j the sum of any two forms is bent, and the forms of $G_0 \cup \cdots \cup G_5$ are pairwise distinct;
- (H3) every $g \in G_j$ has $\langle g, q_j \rangle_4 = \langle g, q_{j+1} \rangle_4 = 1$.

Let $\mathcal{S}_k$ be the bent family of q_k. The *body* consists of the following 7398 vectors of squared norm 8 in $\mathbb{R}^{16} \oplus \mathbb{R}^2$:

- the 480 roots $(\pm 2e_i \pm 2e_j,\ 0)$;
- for each of the 30 octads, the 128 vectors with entries ± 1 on the octad, an odd number of them negative, and zero tail: 3840 vectors;
- the six poles $(0,\ \sqrt{8}\, u_k)$;
- for $k = 0, \dots, 5$ and each sextad $S \in \mathcal{S}_k$, the 32 vectors with entries ± 1 on S, an even number of them negative, and tail $\sqrt{2}\, u_k$: $6 \cdot 16 \cdot 32 = 3072$ *shoulders*.

The *added points* are the vectors

$$v_{g,\ell,j} = \big((-1)^{g+\ell},\ \sqrt{8/3}\, u_{\theta_j}\big), \qquad g \in G_j,\ \ell \text{ affine}, \qquad |v_{g,\ell,j}|^2 = 16 + \tfrac{8}{3} = \tfrac{56}{3},$$

$32|G_j|$ of them in the sector j. Condition (H3) says that the word $g + \ell$ meets every sextad of the two families bounding its sector in an odd number of points.

Theorem 5.1. *The body and the added points of a hexagon configuration are pairwise compatible. Their normalizations form a kissing configuration of* $7398 + 32\sum_j |G_j|$ *points in* $\mathbb{R}^{18}$.

Proof. Body vectors have squared norm 8, so two of them are compatible when their inner product is at most 4; a body vector and an added vector when it is at most $\frac{1}{2}\sqrt{8 \cdot 56/3} = \sqrt{112/3} \approx 6.11$; two added vectors when it is at most $28/3$. If α is the angle between two tails, the tail inner product is $8\cos\alpha$ for two poles, $4\cos\alpha$ for a pole and a shoulder, $2\cos\alpha$ for two shoulders, $\frac{8}{\sqrt{3}}\cos\alpha$ for a pole and an added point, $\frac{4}{\sqrt{3}}\cos\alpha$ for a shoulder and an added

point, and $\frac{8}{3}\cos\alpha$ for two added points. Between two body vectors, or two added points, α is a multiple of $\pi/3$; between a body vector and an added point it is an odd multiple of $\pi/6$.

Body pairs.

- Two distinct roots have inner product 0, ± 4 or -8. A root and any other body vector have at most two nonzero head coordinates in common, each contributing at most 2, and zero tail product.
- Two octad vectors, or an octad vector and a shoulder, share at most 4 coordinates (two octads meet in 0 or 4 points, an octad and a sextad in 2 or 4), and on a common octad two odd patterns differ in at least two signs. Octad vectors have zero tail, so the inner product is at most 4.
- Two shoulders of the same family have tail product 2. If their sextads differ they share 2 points, and if they coincide the two even patterns differ in at least two signs; the head product is at most 2 in both cases.
- Two shoulders of families $k \neq k'$: if $u_{k'} = -u_k$ and $q_{k'} = q_k$, the tail product is -2 and the head product at most 6. Otherwise $q_k + q_{k'}$ is bent by (H1), the sextads share 1 or 3 points, the head product is at most 3, and the tail product is at most 1.
- A pole and a shoulder give $4\cos\alpha \leq 4$, and two poles $8\cos\alpha \leq 4$.

Body against added. Let $v = v_{g,\ell,j}$.

- An octad vector has an odd number of minus signs on its octad, while $g + \ell \in \mathrm{RM}(2,4)$ meets the octad, a word of the dual code, in an even number of points. So the two sign patterns disagree somewhere on the octad, and the inner product is at most $8 - 2 = 6$.
- A root gives at most 4, and a pole at most $\frac{8}{\sqrt{3}}\cos\frac{\pi}{6} = 4$.
- A shoulder on a sextad S of the family k: if $k \in \{j, j+1\}$, then $\alpha = \pi/6$, the tail product is 2, and by (H3) the word $g + \ell$ meets S in an odd number of points, so the two patterns on S, one even and one odd, disagree somewhere and the head product is at most $6 - 2 = 4$; the sum is at most 6. If $\alpha = \pi/2$ the tail product is 0 and the head product at most 6. Otherwise the tail product is negative.

All cases are below $\sqrt{112/3}$.

Two added points. In one sector the heads are distinct words of the cosets $g + \mathrm{RM}(1,4)$, $g \in G_j$. Two words of one coset differ by an affine word of weight 8 or 16, and two words of different cosets differ by a word of weight 6 or 10, because the sum of the two forms is bent by (H2). So the head product is at most $16 - 12 = 4$, the tail product is $\frac{8}{3}$, and the sum is at most $\frac{20}{3}$. In different sectors the heads are distinct words of $\mathrm{RM}(2,4)$, because the forms are distinct by (H2) and distinct forms lie in different cosets of $\mathrm{RM}(1,4)$; so the head product is at most $16 - 8 = 8$, the tail product is at most $\frac{8}{3}\cos\frac{\pi}{3} = \frac{4}{3}$, and the sum is at most $\frac{28}{3}$. □

The section is a hexagon configuration. We return to the coordinates of Section 3. Let F be the set of the 30 octads of $D^\perp$ disjoint from T, the octads of the full fibers.

Lemma 5.2. *$F \cup \{\emptyset, H\}$ is a linear code of length 16 on the coordinates H. The coordinates can be identified with $\mathbb{F}_2^4$ so that this code is $\mathrm{RM}(1,4)$ and F is its set of octads.*

Proof. The octads of F are the octads of G disjoint from $O = T \cup P$. The sum of two of them is a word of G supported on H, of weight 8 or 16, so it is again in F or equals H; and H itself is in G, as the complement of O. So $C = F \cup \{\emptyset, H\}$ is a code of dimension 5 whose nonzero weights are 8 and 16, with the weight enumerator $1 + 30y^8 + y^{16}$ of $\mathrm{RM}(1,4)$. By the MacWilliams identity $C^\perp$ has the weight enumerator of $\mathrm{RM}(2,4)$, so its minimum distance is 4. Hence no column of a generator matrix of C is zero and no two are equal. Take a basis $\mathbf{1}_H, c_1, \ldots, c_4$

of C and map the coordinate $i \in H$ to $(c_1(i), \dots, c_4(i)) \in \mathbb{F}_2^4$. The map is injective, hence a bijection, and under it the words of C are exactly the affine functions $\sum a_k c_k + b$. □

For the rest of the section fix this identification. We rotate the tail plane $s^\perp \cap \mathbb{R}^T$ onto $\mathbb{R}^2$ so that the six vectors $(e_i - e_j)/\sqrt{2}$, $i \neq j \in T$, become $u_0, \dots, u_5$; for each pair $\{i, j\} \subset T$ the two directions $\pm(e_i - e_j)$ are opposite. By Lemma 4.1(1) the six tails $t(c)$ then lie at the midpoint angles.

Proposition 5.3. *The configuration of Theorem 3.3 is a hexagon configuration.*

(1) *For each pair $\{i, j\} \subset T$, the sixteen sets $o \setminus T$, where o runs over the octads of $D^\perp$ with $o \cap T = \{i, j\}$, are the sextads of a bent form q_{ij}; the three forms have pairwise bent sums. The half-fiber vectors on these octads are the shoulders of the two directions $\pm(e_i - e_j)$, both with the family of q_{ij}, and the tail roots are the poles. So (H1) holds with $q_{k+3} = q_k$.*

(2) *For each of the six tail values, the 160 heads $q(c)_H$ of the words $c \in A_1$ with that tail value form five cosets of* RM$(1, 4)$ *in* RM$(2, 4)$. *The five forms satisfy (H3), and the thirty forms are pairwise distinct. The sum of two forms of one sector is bent (a finite check), so (H2) holds.*

Proof. (1) Two octads o, o' with $o \cap T = o' \cap T = \{i, j\}$ meet in 2 or 4 points. If they met only in $\{i, j\}$, their sum would be a dodecad of G disjoint from the octad O, but a dodecad meets every octad in 2, 4 or 6 points [3, Ch. 10, §2]. So o and o' meet in 4 points, the sets $o \setminus T$ and $o' \setminus T$ meet in 2, and their sum $o + o'$ is an octad of F. Hence the sixteen sets lie in one coset of RM$(1, 4)$, which contains them and their complements: 32 words of weight 6 and 10. The coset is $q_{ij} +$ RM$(1, 4)$ for a Boolean function q_{ij} all of whose translates have weight 6 or 10, that is, a bent function, and bent functions on $\mathbb{F}_2^4$ are quadratic [8, Ch. 14]. Two octads o, o' with $o \cap T = \{i, j\}$ and $o' \cap T = \{j, k\}$ meet in 2 or 4 points, one of them j, so $o \setminus T$ and $o' \setminus T$ meet in 1 or 3 points, their sum has weight 10 or 6, and $q_{ij} + q_{jk}$ is bent. The half-fiber vectors on o have an even number of minus signs on $o \setminus T$ and tail $\pm(e_i - e_j)$, of squared norm 2 (proof of Lemma 3.1), and the tail roots $2(e_i - e_j)$ have squared norm 8.

(2) By the proof of Lemma 3.2, the words of A_1 with a given tail value are the words of restriction τ in the five cosets $M + a_k + r$ for the one $r \in \{r_1, r_2, r_1 + r_2\}$ with $r \cap T \in \{\tau, \tau + T\}$; in each coset they form a coset of $M_0 = \{m \in M : m \cap T = \emptyset\}$. Now M_0 is spanned by m_1, m_3, m_4, m_5, m_6, which are octads of F, so M_0 restricted to H is RM$(1, 4)$ by Lemma 5.2, and the 160 heads form five cosets of RM$(1, 4)$; they are cosets in RM$(2, 4)$ because D restricted to H is orthogonal to F. For (H3): a word $c \in D$ meets every octad o of $D^\perp$ evenly, so for $o \cap T = \{i, j\}$ the head c_H meets the sextad $o \setminus T$ in $|c \cap o| - |c \cap \{i, j\}|$ points, which is odd exactly when $|c \cap \{i, j\}|$ is odd. The tail $t(c)$ lies at the midpoint of the two directions $\pm(e_i - e_j)$ with $|c \cap \{i, j\}|$ odd and is perpendicular to the third: for $c \cap T = \{14\}$, $t(c) = \frac{1}{3}(2, 2, -4)$ has inner product 2 with $e_2 - e_{14}$ and $e_{11} - e_{14}$ and 0 with $e_2 - e_{11}$, and the other five cases are permutations and negations. So every head meets every sextad of the two families bounding its sector oddly. Finally, the thirty forms are distinct: if a form g served two sectors, the two added points with the head $(-1)^g$ would have inner product at least $16 - \frac{8}{3} > \frac{28}{3}$, contradicting Theorem 3.3. □

So the section of Ho's configuration and the construction of the next section are two ways of filling one shape. The compatibility of the added points of the section was proved in Theorem 3.3 from the distance 5 of Ho's code; Theorem 5.1 proves it again from the bent sums. In both constructions five forms per sector is the most the shape allows: for the families

of the section, and for those of Section 6, a search over the 64 quadratic forms finds no six forms with pairwise bent sums that satisfy (H3).

6. A second construction: six bent families on a hexagon

The section uses three bent families, each for two opposite directions. We now take six different forms, with all pairwise sums bent, and fill the six sectors again with five forms each. This configuration of 8358 points is built directly in $\mathbb{R}^{16} \oplus \mathbb{R}^2$ and does not pass through 19 dimensions; Section 7 shows that it is not isometric to the section. Take

$$q_0 = x_0x_3 + x_1x_2, \qquad q_1 = x_0x_2 + x_0x_3 + x_1x_3,$$
$$q_2 = x_0x_1 + x_0x_2 + x_1x_2 + x_1x_3, \qquad q_3 = x_0x_1 + x_0x_2 + x_0x_3 + x_2x_3,$$
$$q_4 = x_0x_1 + x_1x_2 + x_2x_3, \qquad q_5 = x_0x_1 + x_0x_3 + x_1x_3 + x_2x_3,$$

whose pairwise sums are bent, so that (H1) holds, and let G_j be the five forms of row j of Table 2. The table was found by computer search. Its three properties, each a short check on the polynomials, are (H2) and (H3): the thirty forms are distinct; in each row, the sum of any two forms is bent; and each form g in row j has $\langle g, q_j\rangle_4 = \langle g, q_{j+1}\rangle_4 = 1$.

Table 2. The quadratic forms of the six sectors. Row j lists the five forms of G_j, used at the angle $\theta_j = (j + \frac{1}{2})\pi/3$ between the families of q_j and q_{j+1}. A monomial x_ix_j is written ij.

j	forms of G_j				
0	$02 + 03$	12	$03 + 13$	$01 + 02 + 12 + 13$	$02 + 12 + 13 + 23$
1	02	$03 + 12$	13	$01 + 02 + 03 + 12 + 13$	$01 + 12 + 23$
2	$01 + 03$	$02 + 12$	$01 + 12 + 13$	$02 + 03 + 23$	$03 + 12 + 13 + 23$
3	01	$02 + 03 + 12$	$02 + 23$	$01 + 02 + 03 + 12 + 23$	$12 + 13 + 23$
4	$01 + 02 + 12$	$01 + 13$	$03 + 12 + 13$	23	$01 + 03 + 12 + 13 + 23$
5	$01 + 03 + 13$	$12 + 13$	$03 + 23$	$02 + 12 + 23$	$01 + 12 + 13 + 23$

Theorem 6.1. *The hexagon configuration with these forms is a kissing configuration of* $7398 + 6 \cdot 5 \cdot 32 = 8358$ *points in* $\mathbb{R}^{18}$.

Proof. Conditions (H1) to (H3) hold, so Theorem 5.1 applies. □

This proves the second half of Theorem 1.1, apart from the non-isometry. The configuration has 1 364 870 contacts, 1 352 838 among body points and 12 032 among added points in different sectors (computer count); no added point touches a body point.

7. Comparison of the configurations

Proposition 7.1. *The configuration of Theorem 3.3 and that of Theorem 6.1 are not isometric. The first contains the antipodes of at least 7398 of its points, the second of exactly 4326.*

Proof. The number of points whose antipode is also in the configuration is invariant under orthogonal maps. In the section of Ho's configuration the body is antipodal: negating a root gives a root, and negating an octad vector changes eight signs and keeps the parity. So at least 7398 points have their antipodes present. In the six-family configuration the roots, the octad vectors and the six poles are antipodal, which gives $480 + 3840 + 6 = 4326$ points. No shoulder has its antipode present: the antipode has tail $-\sqrt{2}u_k = \sqrt{2}u_{k+3}$, so it would be a shoulder on a sextad of q_{k+3}, but sextads of q_k and q_{k+3} meet in 1 or 3 points and are never equal. No

added point has its antipode present: the antipode of $v_{g,\ell,j}$ has head $(-1)^{g+\ell+1}$, a word of the coset $g + \mathrm{RM}(1,4)$, and tail at the angle θ_{j+3}, whose sector uses five forms different from g and hence five other cosets. □

In fact the section contains the antipodes of exactly 7398 of its points, by the argument at the end of Remark 4.5 with $\mu_c = 1$: no added point has its antipode present, because no two words of A_1 are complementary (computer check).

Comparison with Kravatskiy's configuration. Kravatskiy's repository [5] describes its 8358 points by a recipe and data files rather than coordinates. In the language of Section 5 it is a hexagon configuration of the same kind as the section: three bent families, each serving two opposite directions, and 960 added points in the six sectors. It differs from the section in its sign conventions, which are the reverse of ours (its shoulders have an odd number of minus signs and its added words meet the neighbouring sextads evenly), and in the scale of its added points. With head entries ± 1, its added points have tail length $\sqrt{2}$ and squared norm 18, so that they touch the octad vectors, where ours have tail length $\sqrt{8/3}$, squared norm $56/3$, and touch nothing in the body.

We rebuilt Kravatskiy's configuration from the data in the repository and computed its invariants exactly. It has 1 621 638 contacts and contains the antipodes of 7398 of its points. The section has 1 367 430 contacts and the six-family configuration 1 364 870. Since the number of contacts is invariant under isometry, neither of our configurations is isometric to Kravatskiy's, and by Proposition 7.1 they are not isometric to each other. The three configurations of 8358 points fill one shape in three different ways, and they are pairwise different.

8. Verification

The three configurations of Section 7 are given as data. The configuration of Theorem 3.3 is given as 8358 integer vectors in $\mathbb{R}^{19}$: the 7398 body vectors of squared norm 8 and the 960 vectors $y(c)$ of squared norm 168. Every vector satisfies $x_2 + x_{11} + x_{14} = 0$, and inner products are those of $\mathbb{R}^{19}$, so a check in these coordinates is the 18-dimensional check. The configuration of Theorem 6.1 is given as 8358 vectors in $\mathbb{R}^{18}$ with coordinates in $\mathbb{Z}[\sqrt{2}, \sqrt{3}]$ (body vectors scaled by 2 and added vectors by 3), and our rebuild of Kravatskiy's configuration as 8358 vectors with coordinates in the same ring.

A program checks all $\binom{8358}{2} = 34\,923\,903$ pairs of each list in exact arithmetic, with no symmetry reduction and no floating point: for every pair with $\langle x, y\rangle > 0$ it decides the sign of $4\langle x, y\rangle^2 - |x|^2|y|^2$ exactly, in integers or in $\mathbb{Z}[\sqrt{2}, \sqrt{3}]$. It finds no violation and no two vectors with the same direction in any of the three lists. For the first configuration an independent second program agrees on all pairs and computes the rank, 18, exactly. These checks use none of the lemmas above.

The computed facts used in the paper are Lemma 4.2; the bent sums in Proposition 5.3(2) and the maximality of five forms per sector; the contact counts and the antipode count of the rebuilt configuration in Section 7; and the checks named in Remark 4.5.

AI usage statement

GPT-6 Astra Pro, Claude Opus 5.5 and Claude Fable 5.1 were used to find these constructions, prepare supplementary data and draft this paper. The manuscript was proofread and lightly revised by the author. The author welcomes detailed questions regarding the use of AI in this work.

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Data availability

The coordinates of the configurations of Theorems 3.3 and 6.1, our rebuild of Kravatskiy's configuration, and an exact checking program accompany this paper. They can be found at `https://github.com/Felpix-Studios/kissing-number`.

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